

Customer Churn Intelligence

Overview of churn patterns, customer risk and model performance

Churn Overview

Churn Drivers

Customer Risk

Total Customers

7,043

Actual Churn Rate

26,5 %

High-Risk Customers

3,235

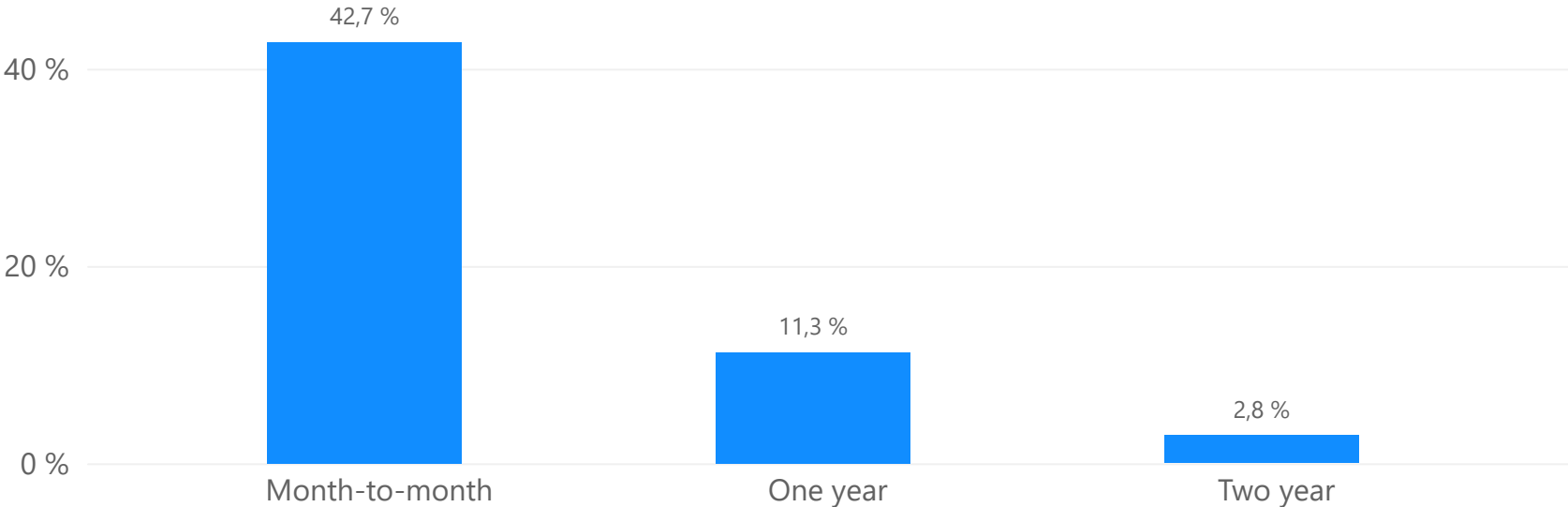
Average Churn Probability

39,3 %

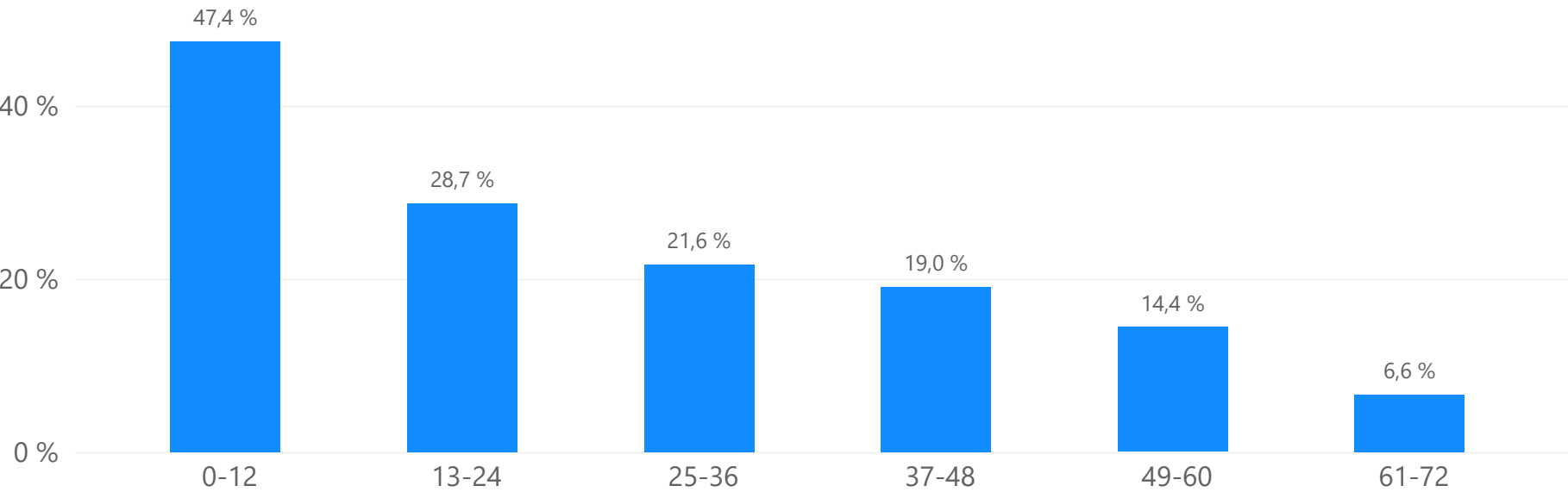
Model performance — held-out test set

Accuracy: 76.7% | Recall: 71.9% | F1: 62.1% | ROC-AUC: 84.2%

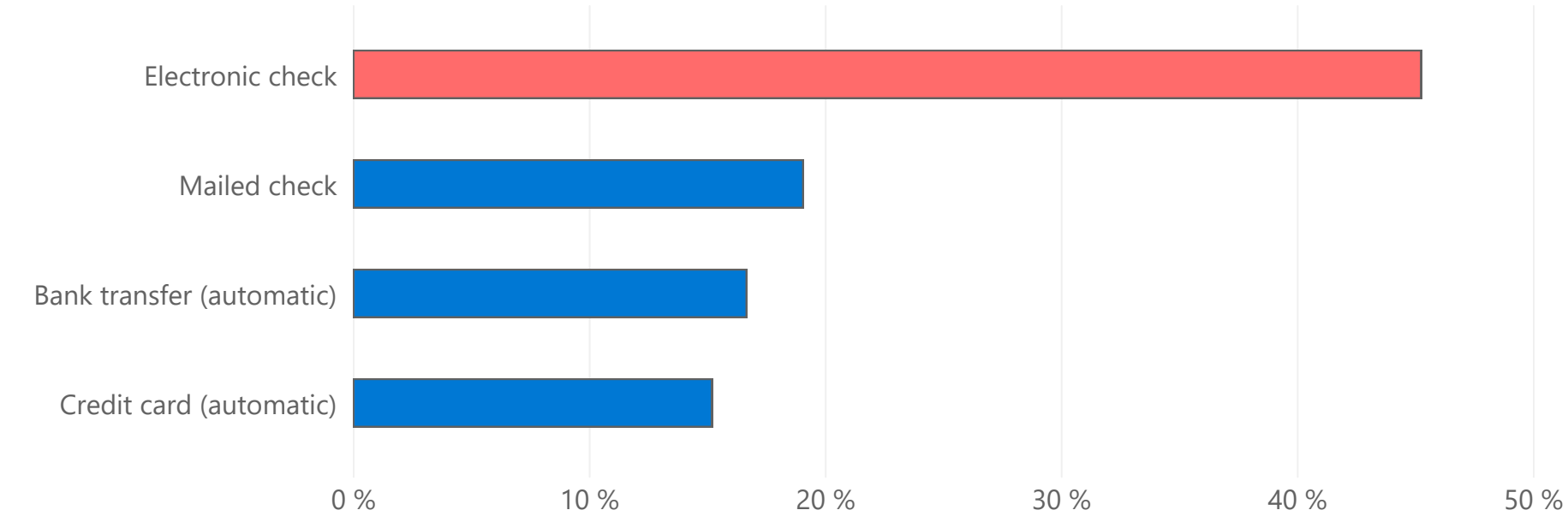
Churn Rate by Contract Type



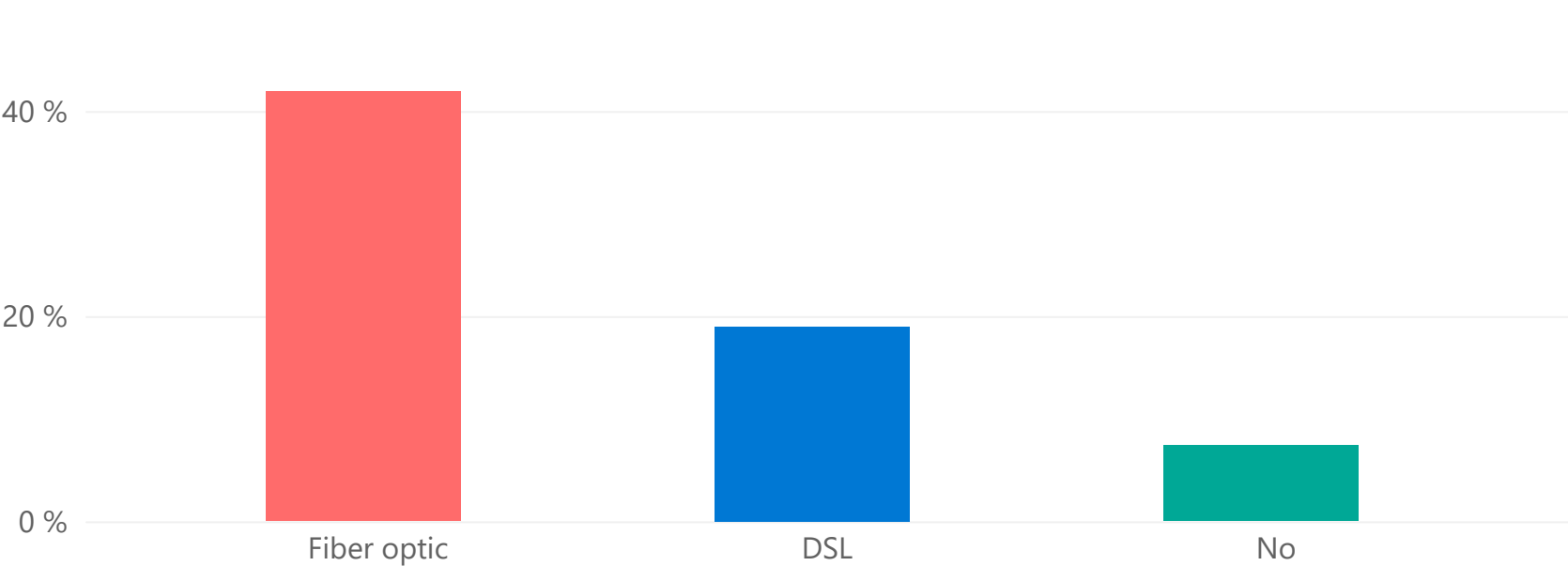
Churn Rate by Tenure Group



Churn Rate by Payment Method



Churn Rate by Internet Service



Churn Drivers

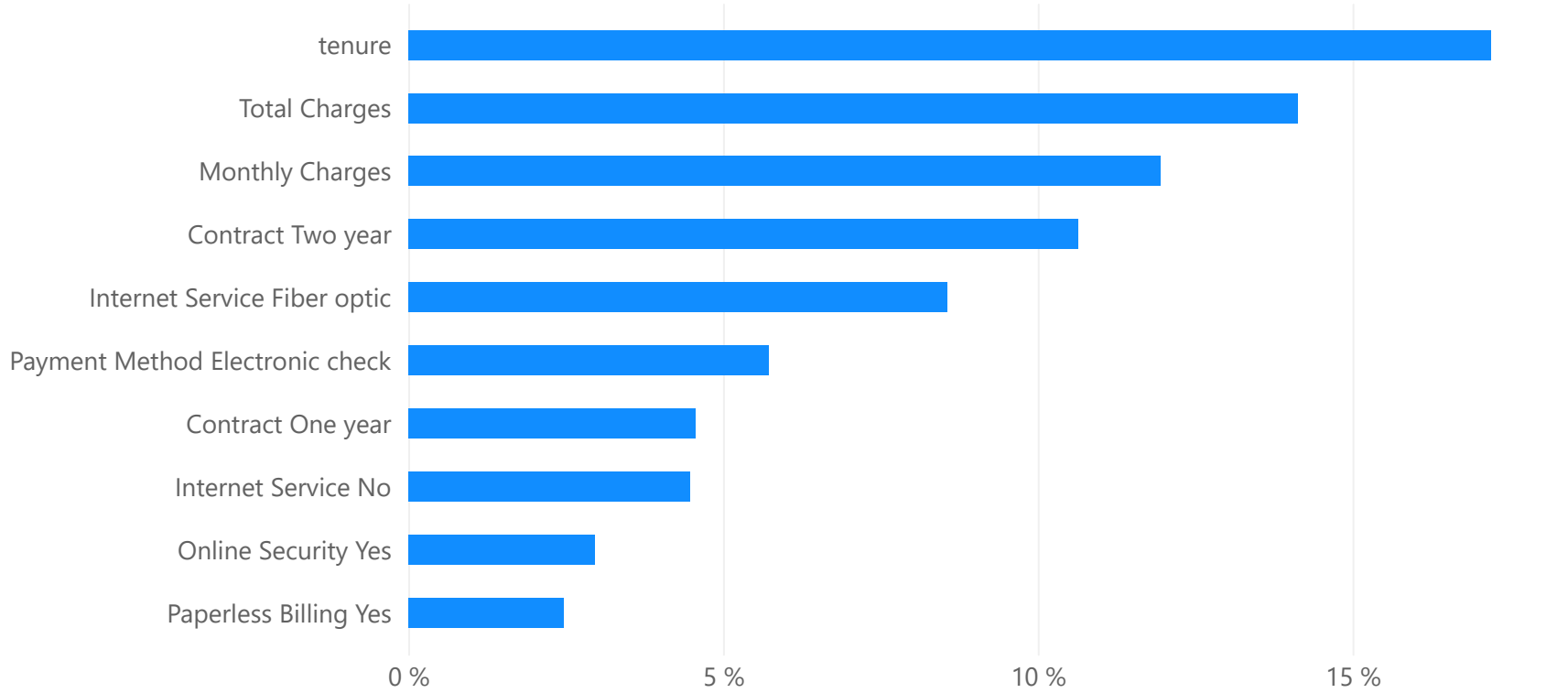
Features most associated with churn in the predictive models

Churn Overview

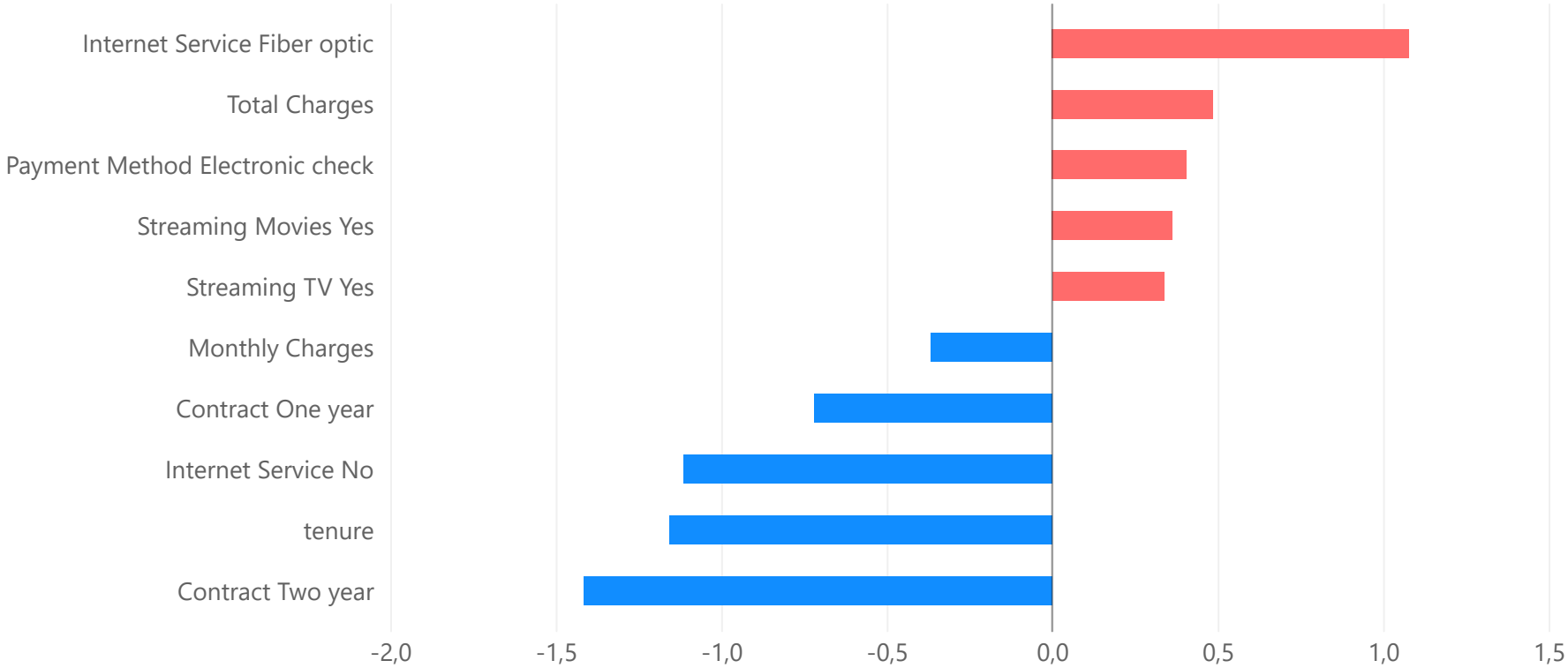
Churn Drivers

Customer Risk

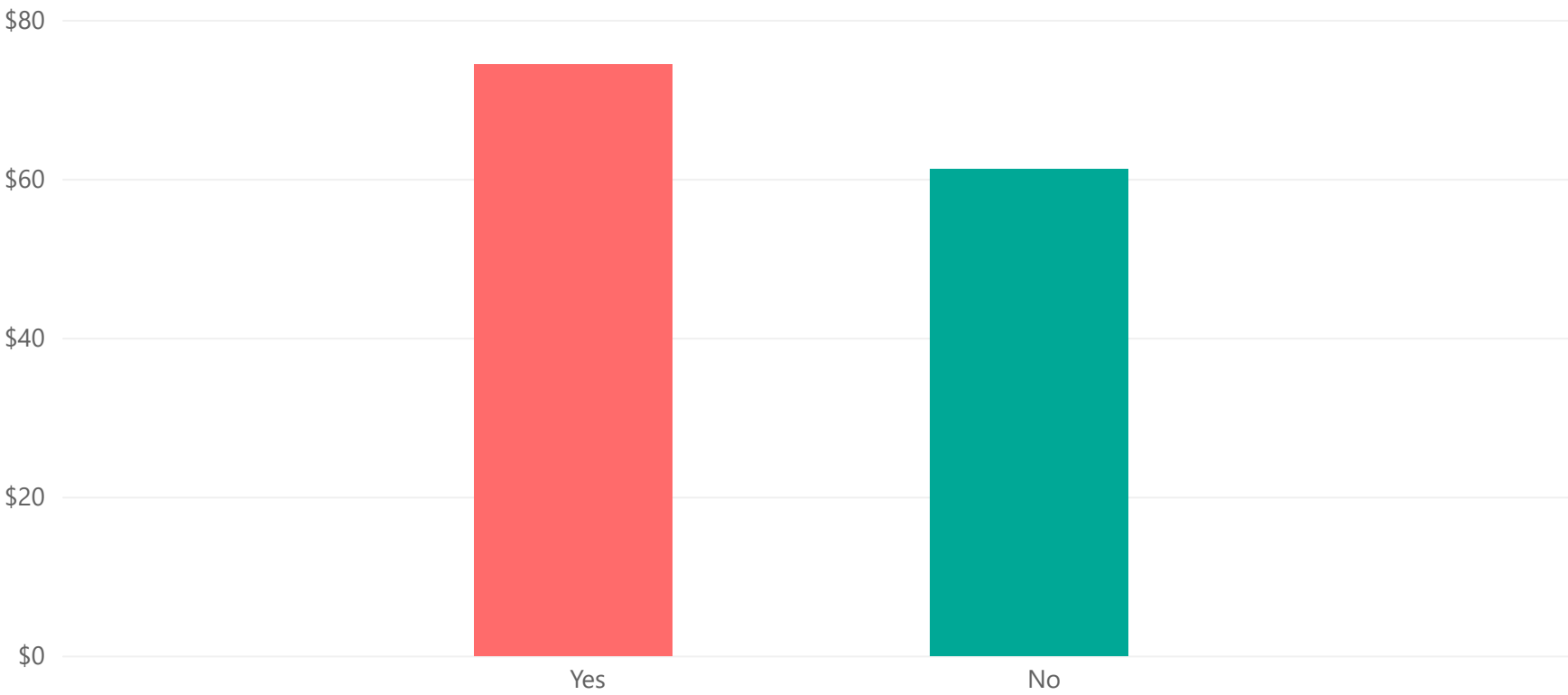
Random Forest — Feature Importance



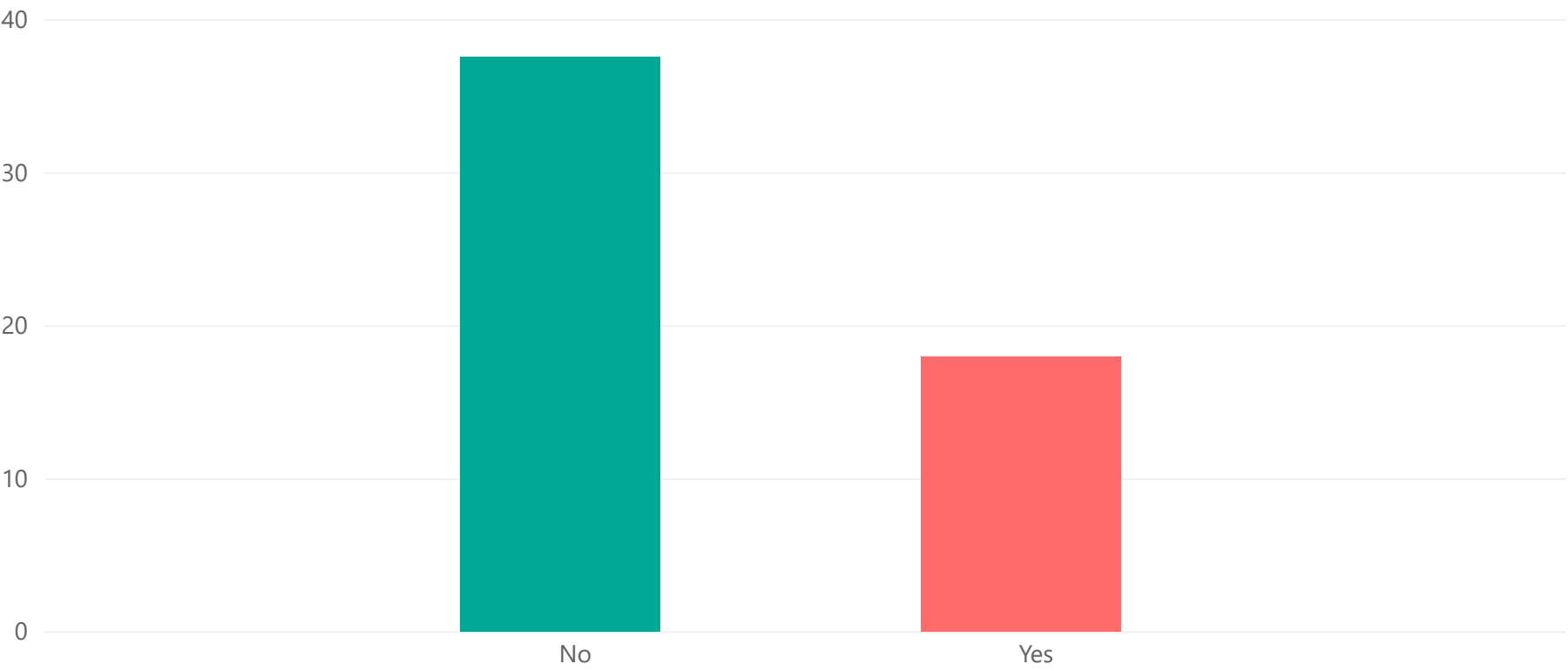
Logistic Regression — Features Associated With Churn



Average Monthly Charges — Churned vs Retained



Average Tenure — Churned vs Retained



Customer Risk

Customers classified as high risk based on predicted churn probability

High-Risk Customers

3,235

Average Churn Probability

68,6 %

High Risk Threshold

≥ 40%

Contract

☐ Month-to-month

☐ One year

☐ Two year

Internet Service

☐ DSL

☐ Fiber optic

☐ No

Payment Method

Bank transfer (automatic)

Credit card (automatic)

Electronic check

Mailed check

Tenure Group

0

72

Churn Probability

40,0 %

95,8 %

Customer	Churn Probability		Contract	Tenure (months)	Internet Service	Payment Method	Monthly Charges
7216-EWTRS		95,8 %	Month-to-month	1,00	Fiber optic	Electronic check	\$100,80
5178-LMXOP		95,7 %	Month-to-month	1,00	Fiber optic	Electronic check	\$95,10
9300-AGZNL		95,6 %	Month-to-month	1,00	Fiber optic	Electronic check	\$94,00
9497-QCMMS		95,6 %	Month-to-month	1,00	Fiber optic	Electronic check	\$93,55
5419-JPRRN		95,5 %	Month-to-month	1,00	Fiber optic	Electronic check	\$101,45
0295-PPHDO		95,2 %	Month-to-month	1,00	Fiber optic	Electronic check	\$95,45
1069-XAIEM		95,0 %	Month-to-month	1,00	Fiber optic	Electronic check	\$85,05
0107-YHINA		94,8 %	Month-to-month	1,00	Fiber optic	Electronic check	\$99,75
4910-GMJOT		94,8 %	Month-to-month	1,00	Fiber optic	Electronic check	\$94,60
8149-RSOUN		94,7 %	Month-to-month	1,00	Fiber optic	Electronic check	\$93,85
9248-OJYKK		94,7 %	Month-to-month	1,00	Fiber optic	Electronic check	\$76,45
4826-XTSOH		94,6 %	Month-to-month	1,00	Fiber optic	Electronic check	\$86,05
3068-OMWZA		94,3 %	Month-to-month	1,00	Fiber optic	Electronic check	\$88,80
4988-IQIGL		94,3 %	Month-to-month	1,00	Fiber optic	Electronic check	\$75,35
6894-LFHLY		94,3 %	Month-to-month	1,00	Fiber optic	Electronic check	\$75,75
5192-EBGOV		94,2 %	Month-to-month	1,00	Fiber optic	Electronic check	\$85,70
9282-IZGQK		94,2 %	Month-to-month	1,00	Fiber optic	Electronic check	\$85,00
8603-IJWDN		94,2 %	Month-to-month	1,00	Fiber optic	Electronic check	\$86,60
3551-HUAZH		94,1 %	Month-to-month	1,00	Fiber optic	Electronic check	\$74,40
4550-EVXNY		94,1 %	Month-to-month	1,00	Fiber optic	Electronic check	\$69,65
8779-QRDMV		94,1 %	Month-to-month	1,00	DSL	Electronic check	\$39,65
Avg		68,6 %			18,70		\$74,53